"Functions & Limits"

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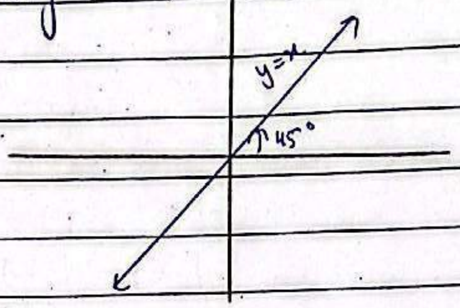
Identity Function:

$y = I(x) = x \quad \forall\, x \in$ Domain. $\quad I : X \to X$ (Domain → Range.)

↳ Domain of I = Range of I

↳ $f \circ f^{-1}(x) = f^{-1} \circ f(x) = x$

↳ Graph = Line bisecting first & 3rd Quad.



Constant Function:

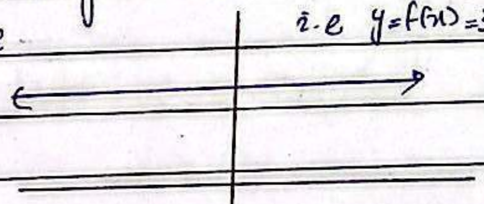
$y = f(x) = c \; ; \; c \in \mathbb{R}$

Range $(f) = \{c\}$

↳ Range is Singleton Set.

↳ constant function is many-to-one function.

↳ Graph is a horizontal line. i.e $y = f(x) = 3 \; \forall x \in \mathbb{R}$



Absolute value Function (Modulus Function):

piece wise function.

$$f(x) = |x| = \begin{cases} -x & ; \; x < 0 \\ +x & ; \; x \geq 0 \end{cases}$$

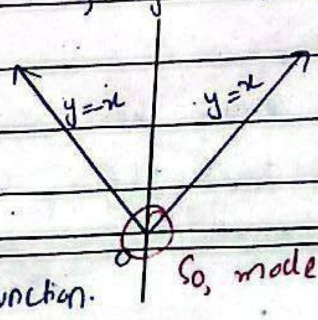
Note: $|x| = \pm x$

(i) $x < 0 \; ; \; |x| = -x > 0$

(ii) $x \geq 0 \; ; \; |x| = x > 0$

$x < 0$, $-x > 0$] inequality changes by multiplying with minus.

(i) Domain $= \mathbb{R}$, Range $[0, +\infty)$



Jahan kisi function ka corner ban jaye wahan wo function diffrenciable nahi hota hai.

So, mode of x is not diffrenciable at $\boxed{x=0}$

(ii) $y = |x|$ is an even function.

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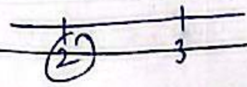
Bracket Function / Greatest integer Function

$f(x) = [x]$ = greatest integer but not greater than x

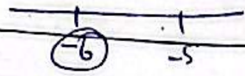
e.g $[3.8] = 3$

$[2.1] = 2$

not a integer



$[-5.8] = -6$



$[3] = [3]$

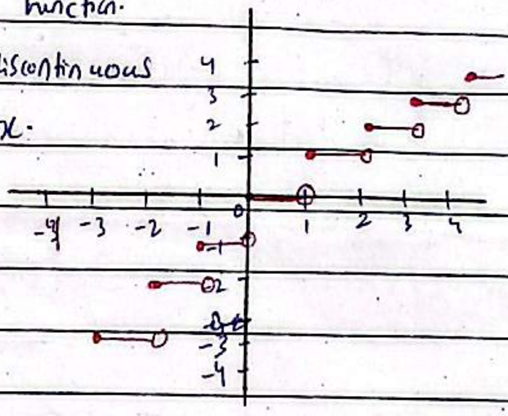
$[-2] = [-2]$

Properties :-

(i) Domain = $\mathbb{R}$, Range = $\mathbb{Z}$

(ii) Graph : unit Step function.

(iii) Bracket function is discontinuous at any integral value of x.



Explicit Function : if y is easily expressed in terms of the independent variable x.

can be a relation also.

e.g $xy + 2x^2 - 1 = 0$

$y = \dfrac{1 - 2x^2}{x}$

we write it as "$y = f(x)$"

Impossible. **Implicit Function :** if y cannot be expressed in terms of the independent variable x.

$xy + y^2 + x = 0 \Rightarrow y \neq f(x)$

we write it as $f(x, y) = 0$

Even function is not a one-one function.

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PARAMETRIC Function: if x & y are expressed in terms of a third variable (i.e Parameter)

$$x = at^2, \quad y = 2at$$

$$x = f(t) \; ; \; y = g(t)$$

So, $\left.\begin{matrix} x = at^2 \\ y = 2at \end{matrix}\right\}$ Parametric Function.

if you eliminate t, Aik equation se t ki value nikal kar dosri mai put kar dein to ye dono equation represent karyein gein aik Simple relationship ko & that relationship is called.

↳ General Equation:

$$\boxed{y^2 = 4ax}$$

Note:- Every even function is Symmetric about $\boxed{\text{y-axis}}$ & every odd function is Symmetric about $\boxed{\text{origin.}}$

↳ Sum of 2 even functions is also even function.

↳ Sum of 2 odd function is also odd.

↳ Product of even & odd function is odd function.

Types of intervals:-

$[a,b]$ = closed interval

$x \in [a,b] \Rightarrow a \le x \le b$

$]a,b[= (a,b)$ = open interval.

$x \in]a,b[\Rightarrow a < x < b$

$[a,b[$ = closed open interval.

$x \in [a,b[\Rightarrow a \le x < b$

$]a,b]$ = open closed interval.

$x \in]a,b] \Rightarrow a < x \le b$

Polynomial functions :- power whole function.

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(i) Dom(polynomial function) = $\mathbb{R} = (-\infty, +\infty)$

(ii) Range (Linear polynomial) = $\mathbb{R} = (-\infty, +\infty)$

Square root function:-

$f(x) = \sqrt{x-1}$

(i) Domain, Put $x - 1 \geq 0$

$x \geq 1$

So, Dom(f) = $[1, +\infty)$

(ii) Range: As, f(x) is a Square root function $= [0, +\infty)$.

$\sqrt{9} = 3$

$x^2 = 9$

$\sqrt{x^2} = \sqrt{9}$

$|x| = 3$

$\pm x = 3$

$x = \pm 3$

$\sqrt{}$

function of function khabi bi aik input par 2 output nahi deta hai.

$f(x) = \sqrt{x}$

$\sqrt{9} = \pm 3$

$= (9, 3)(9, -3)$

not a function.

$\sin^2 x + \cos^2 x = 1$

But $\cosh^2 x - \sinh^2 x = 1$

$1 + \tan^2 x = \sec^2 x$

But $1 - \tan^2 hx = \sec^2 hx$

$1 + \cot^2 x = \operatorname{cosec}^2 x$

But $\coth^2 x - 1 = \operatorname{cosech}^2 x$

multiplication & division formula don't change.

$\sin(2x) = 2\sin x \cos x$	$\sinh(2x) = 2\sinh x \cosh x$
$\tan x = \sin x / \cos x$	$\tanh(x) = \dfrac{\sinh x}{\cosh x}$.

Composition of function:-

$f: X \longrightarrow Y$ & $g: Y \longrightarrow Z$

$gof: X \longrightarrow Z$

defined as $gof(x) = g(f(x))$

Dom (gof) = X

Range (gof) = Z

fog gof fof gog

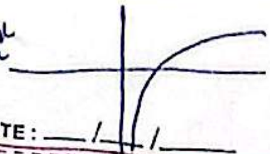
$f(x) = x + 3$; $g(x) = x^2$

$f: \mathbb{R} \rightarrow \mathbb{R}$ $\quad g: \mathbb{R} \longrightarrow [0, +\infty)$

Domain Range

$gof(x) = g(f(x)) = (x+3)^2$

$y=\log_a x$



Log x graph lies in 1 & 4rth Quadrant

$x>0$
$y\in\mathbb{R}$

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Inverse of a Function:

e.g $f(x)=\frac{x-3}{2}$; $f^{-1}(x)=?$

As, $y=f(x)$

So, $y=\frac{x-3}{2}$

$2y=x-3$

$x=2y+3$

$\Rightarrow f^{-1}(y)=x$

$f^{-1}(y)=2y+3$

$\boxed{f^{-1}(x)=2x+3}$

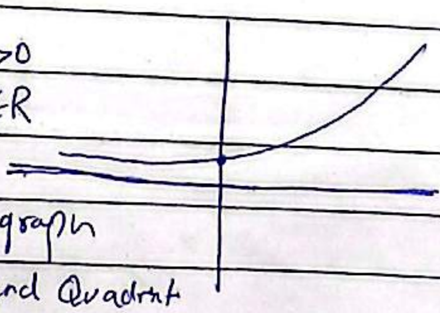
Note:- $f\circ f^{-1}(x)=f^{-1}\circ f(x)=x$ = Identity Function.

$\sin(\sin^{-1}x)=\sin^{-1}(\sin x)=x$

Note:- ~~Domf~~ $\Rightarrow$ $\text{Dom}(f)=\text{Range}(f^{-1})$

$\text{Range}(f)=\text{Dom}(f^{-1})$

$y=a^x$ $y>0$
$x\in\mathbb{R}$



Exponential function graph lies in 1st & 2nd Quadrant

~~Cosh~~

$\cosh^{-1}x=\ln(x+\sqrt{x^2-1})$

$\sinh^{-1}x=\ln(x+\sqrt{x^2+1})$

$\tanh^{-1}x=\frac{1}{2}\ln\frac{1+x}{1-x}$ $|x|<1$

$\cosh x=\frac{e^x+e^{-x}}{2}$

$\sinh x=\frac{e^x-e^{-x}}{2}$

$\tanh x=\frac{e^x-e^{-x}}{e^x+e^{-x}}$

For a rational function $\frac{P(x)}{Q(x)}$.

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↳ horizontatal asymptote $y = 0$

if degree of N = degree of D

↳ horizontal asymptote = $\dfrac{\text{leading coefficient of } N}{\text{leading coefficient of } D}$

if degree of N > degree of D

↳ no horizontal asymptote.

For vertical Asymptote

when denominator = 0